Cereals K-Means (6710503739)

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1 Customer Rating of Breakfast Cereals — K-Means

1.1 Data preprocessing

[1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

pd.set_option("display.width", 88)
raw = pd.read_excel("Cereals_Dataset_63_1.xls").iloc[:, :12]
missing = raw.replace(-1, np.nan).isna().any(axis=1)
print(raw.shape)
print(raw.loc[missing, ["Cereals", "carbo", "sugars", "potass"]].
 ↪to_string(index=False))
```

```
(77, 12)
               Cereals  carbo  sugars  potass
        Almond_Delight   14.0       8     NaN
Cream_of_Wheat_(Quick)   21.0       0     NaN
        Quaker_Oatmeal   -1.0      -1   110.0
```

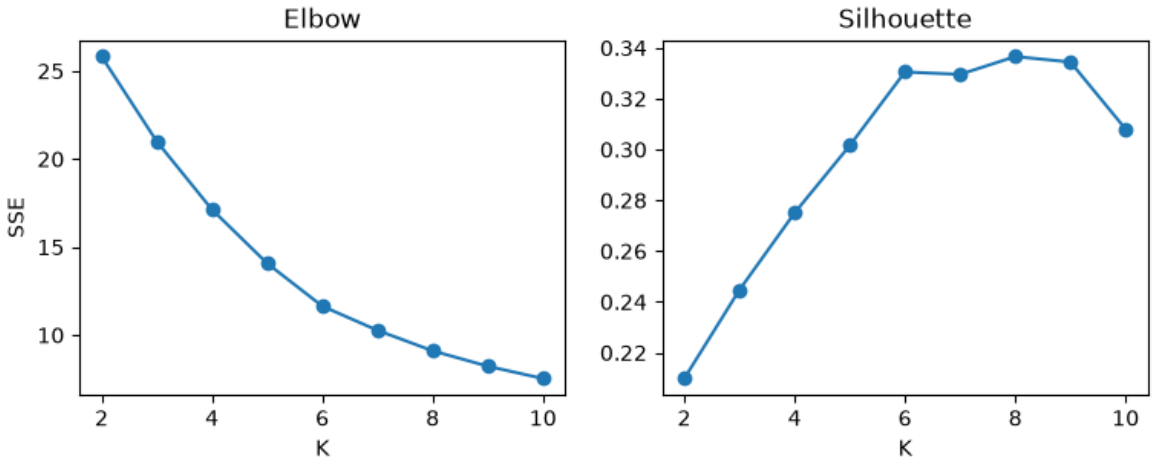
[2]:
```
df = raw[~missing].reset_index(drop=True)
features = ["calories", "protein", "fat", "sodium", "fiber", "carbo", "sugars",␣
 ↪"potass", "vitamins"]
Z = MinMaxScaler().fit_transform(df[features])
print(df.shape)
```

```
(74, 12)
```

1.2 a) K-Means with best K

[8]:
```
ks = range(2, 11)
inertia, sil = [], []
for k in ks:
    km = KMeans(n_clusters=k, n_init=50, random_state=42).fit(Z)
    inertia.append(km.inertia_)
    sil.append(silhouette_score(Z, km.labels_))
```

```
fig, ax = plt.subplots(1, 2, figsize=(9, 3))
ax[0].plot(ks, inertia, "o-"); ax[0].set(title="Elbow", xlabel="K", ylabel="SSE")
ax[1].plot(ks, sil, "o-"); ax[1].set(title="Silhouette", xlabel="K")
plt.show()
```



[4]:
```
BEST_K = 6
df["cluster"] = KMeans(n_clusters=BEST_K, n_init=50, random_state=42).
  ↪fit_predict(Z)
profile = df.groupby("cluster")[features + ["rating"]].mean().round(1)
profile.insert(0, "n", df["cluster"].value_counts().sort_index())
profile.T
```

[4]:

cluster	0	1	2	3	4	5
n	18.0	21.0	17.0	9.0	6.0	3.0
calories	103.3	111.0	123.5	82.2	116.7	63.3
protein	2.7	1.5	3.2	2.4	2.7	4.0
fat	0.5	1.0	2.1	0.1	0.8	0.7
sodium	226.7	168.6	152.9	1.7	208.3	176.7
fiber	1.8	0.6	3.2	2.1	1.8	11.0
carbo	18.3	12.4	13.7	15.3	18.2	6.7
sugars	3.4	11.5	9.1	2.3	6.3	3.7
potass	77.2	47.4	152.1	90.6	95.8	310.0
vitamins	25.0	25.0	23.5	11.1	100.0	25.0
rating	47.4	28.7	39.1	63.0	38.1	73.8

[7]:
```
for k, g in df.groupby("cluster"):
    print(f"c {k}:", ", ".join(g["Cereals"]))
```

```
c 0: Bran_Chex, Bran_Flakes, Cheerios, Corn_Chex, Corn_Flakes, Crispix,
Double_Chex, Grape_Nuts_Flakes, Grape-Nuts, Kix, Multi-Grain_Cheerios, Nutri-
grain_Wheat, Rice_Chex, Rice_Krispies, Special_K, Triples, Wheat_Chex, Wheaties
c 1: Apple_Cinnamon_Cheerios, Apple_Jacks, Cap'n'Crunch, Cinnamon_Toast_Crunch,
```

```
Cocoa_Puffs, Corn_Pops, Count_Chocula, Crispy_Wheat_&_Raisins, Froot_Loops,
Frosted_Flakes, Fruity_Pebbles, Golden_Crisp, Golden_Grahams, Honey_Graham_Ohs,
Honey_Nut_Cheerios, Honey-comb, Lucky_Charms, Nut&Honey_Crunch, Smacks, Trix,
Wheaties_Honey_Gold
c 2: 100%_Natural_Bran, Basic_4, Clusters, Cracklin'_Oat_Bran,
Fruit_&_Fibre_Dates,_Walnuts,_and_Oats, Fruitful_Bran, Great_Grains_Pecan, Life,
Muesli_Raisins,_Dates,_&_Almonds, Muesli_Raisins,_Peaches,_&_Pecans,
Mueslix_Crispy_Blend, Nutri-Grain_Almond-Raisin, Oatmeal_Raisin_Crisp,
Post_Nat._Raisin_Bran, Quaker_Oat_Squares, Raisin_Bran, Raisin_Nut_Bran
c 3: Frosted_Mini-Wheats, Maypo, Puffed_Rice, Puffed_Wheat, Raisin_Squares,
Shredded_Wheat, Shredded_Wheat_'n'Bran, Shredded_Wheat_spoon_size,
Strawberry_Fruit_Wheats
c 4: Just_Right_Crunchy__Nuggets, Just_Right_Fruit_&_Nut, Product_19,
Total_Corn_Flakes, Total_Raisin_Bran, Total_Whole_Grain
c 5: 100%_Bran, All-Bran, All-Bran_with_Extra_Fiber
```

1.3 b) Meaning of each cluster

Cluster	Description
0	Plain corn/rice/wheat flakes
1	Sugary kids' cereals
2	Granola/muesli/raisin bran
3	Puffed/shredded wheat
4	Fortified cereals
5	Bran cereals

1.4 c) Healthy cereals

The healthy-cereal clusters are 3 and 5: low calories, fat, sugar (and no sodium in cluster 3), high fiber in cluster 5, and the two highest average ratings. Cluster 5 alone has only 3 cereals, too few to offer a different cereal every day, so the school should use clusters 3 + 5 together (12 cereals).

[6]:
```
healthy = df[df["cluster"].isin([3, 5])]
cols = ["Cereals", "cluster", "calories", "fat", "sodium", "fiber", "sugars",␣
 ↪"rating"]
print(healthy[cols].sort_values("rating", ascending=False).round(1).
 ↪to_string(index=False))
```

```
                  Cereals  cluster  calories  fat  sodium  fiber  sugars  rating
All-Bran_with_Extra_Fiber        5        50    0     140   14.0       0    93.7
   Shredded_Wheat_'n'Bran        3        90    0       0    4.0       0    74.5
Shredded_Wheat_spoon_size        3        90    0       0    3.0       0    72.8
                100%_Bran        5        70    1     130   10.0       6    68.4
           Shredded_Wheat        3        80    0       0    3.0       0    68.2
             Puffed_Wheat        3        50    0       0    1.0       0    63.0
              Puffed_Rice        3        50    0       0    0.0       0    60.8
                 All-Bran        5        70    1     260    9.0       5    59.4
  Strawberry_Fruit_Wheats        3        90    0      15    3.0       5    59.4
      Frosted_Mini-Wheats        3       100    0       0    3.0       7    58.3
```

Raisin_Squares	3	90	0	0	2.0	6	55.3
Maypo	3	100	1	0	0.0	3	54.9